Lightweight Tuberculosis Screening Pipeline for Community-based Active Case Finding

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Abstract*—Tuberculosis (TB) screening is widely done with computer-aided detection (CAD) systems. While TB CAD systems are highly accurate, community-based active case-finding (ACF) interventions face barriers in connectivity, compute, and interpretation. ACF often conducts high-volume screening in remote community locations where compute and connectivity are limited. The TB CAD system then reads each chest X-ray on-site for presumptive TB and outputs probability scores and visual overlays of abnormalities. However, these CAD outputs are primarily designed for radiologist reading, whereas ACF field teams often do not involve a radiologist and on-site readers without radiology training evaluate the CAD results. We present a TB screening pipeline that pairs lightweight classification and object detection models with a natural-language description of model findings. For classification we introduce FlipR, a lightweight symmetry-gated network that achieves comparable performance to the strongest baselines despite using the fewest parameters. The classifier exceeds the World Health Organization's target sensitivity and specificity on the held-out test set, though at an argmax operating point and on a curated benchmark that are not directly comparable to the WHO 2025 product evaluation. For object detection we benchmark lightweight detectors for active and latent TB localization and find that compact off-the-shelf models outperform a larger custom DraxNet backbone on active TB. Findings are rendered as a deterministic, faithful-by-construction natural-language report following radiologic report convention, so the text asserts only what the models detected. The pipeline targets adult screening and is deliberately lightweight for deployment in resource-limited settings. However, the design and evaluation of the system was constrained by limited TB-specific samples in public datasets. Further work remains to evaluate deployment feasibility, detect specific radiographic findings, and evaluate natural language outputs through a radiologist reader study.

***Keywords*—Tuberculosis, Computer-Aided Detection, Chest X-ray, Medical Imaging, Community-based Active Case Finding**

I. Introduction

Tuberculosis (TB) causes more deaths than any other infectious disease worldwide [1]. It is caused by airborne *Mycobacterium tuberculosis* and primarily affects the lungs, presenting as pulmonary TB [3]. Pulmonary TB is typically diagnosed by assessing patient history of symptoms, chest radiography, and bacteriological confirmation. In a clinical setting, the patient will be interviewed for symptom history by a clinician, typically a general practitioner or pulmonologist. The clinician may then request a chest X-ray (CXR) of the patient. A radiologist interprets CXRs for abnormalities indicative of TB and reports their clinical impression in a radiographic report. If patient symptoms or radiographic report indicate TB, the patient is said to have presumptive TB. Bacteriological confirmation is needed to confirm TB, and is done through smear microscopy, culture, or rapid diagnostics such as Xpert MTB/RIF or Truenat. Once confirmed, the patient is said to have bacteriologically confirmed TB. Upon clinical diagnosis of TB, the patient then begins treatment.

Community-based active case finding (ACF) refers to an intervention where teams screen at-risk populations in community sites or through home visits, rather than waiting for symptomatic patients to present at a clinic [2]. Figure 1 shows a representative ACF workflow, where a field team carries out screening and triage on the community site and an off-site laboratory performs confirmation.

A. Computer-aided Detection of Tuberculosis

TB computer-aided detection (CAD) systems are widely used in both clinical settings and ACF. The World Health Organization (WHO) endorsed CAD systems as an alternative to human readers for patients aged 15 years and older. In 2025, WHO approved six commercial products as meeting its TB screening performance standards [7]. In one reader study, CAD assistance improved radiologist performance and enabled non-radiologists to achieve similar performance as radiologists [9]. Several reviews report CAD model accuracy approach WHO targets across different populations [8], [34], [35], [14].

In ACF, WHO positions TB CAD as a screening and triage tool that selects who proceeds to confirmatory testing [6], [2]. Since CAD systems are used in place of a human reader, community-based ACF teams commonly do not include a radiologist [21], [22]. Despite WHO endorsement, however, clinicians across countries currently value CAD as a complement to, not a replacement for, expert reading [33].

B. Challenges in CAD Integration in Community-based ACF

While TB CAD systems are highly accurate, CAD workflow integration, local validation, and human factors are relatively under-studied [10], [8]. ACF, in particular, faces barriers in connectivity, compute, and interpretation:

Connectivity and compute are constrained in community sites. ACF is often conducted in remote community sites, where connectivity may be unstable and laptops are used to run CAD software [21]. Implementation studies in India and Vietnam document how workflow, connectivity, and interpretation conditions shape CAD deployment in the field [31], [32].

CAD overlays are designed for radiologist readers, whereas ACF teams often do not have radiologists. We review the six TB CAD products approved by WHO in 2025: qXR v4.0 [15], CAD4TB v7 [16], Lunit INSIGHT CXR v3.1 [17], DrAid v2.4.4–6 [18], Genki Edge v3.4.2 [19], and

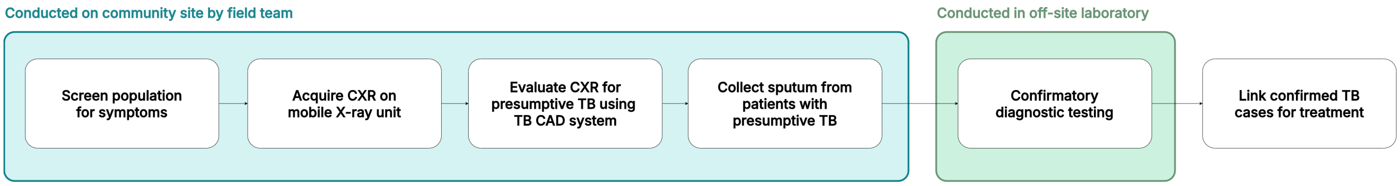

Fig. 1. A TB community active case-finding (ACF) workflow. The CAD system in ACF is often used by a radiographer rather than a radiologist.

InferRead DR Chest v1.0.1.1 [20] [7]. All deliver TB screening results as a probability score with visual localization through a boxed region, outline, or heatmap. Some products additionally generate broader radiology reports spanning many chest findings [18], [15], but such reports are produced for radiologist reading rather than for the non-radiologist readers in ACF, and the TB screening output itself remains a score and an overlay. CAD overlays are also found to overlap poorly with expert-identified regions and do not convey the contextual reasoning used during diagnosis [27]. A TB CAD acceptability study found that some users wanted to understand why the model reached its conclusions [38], and textual descriptions have been proposed as a more clinically meaningful complement to image markup [36], [28], [11], [37]. This presents unique constraints to ACF, which is often carried out by mobile field teams that include a radiographer but often no radiologist on-site [22]. Despite chest X-ray interpretation falling outside the radiographer's scope of practice, the radiographer in ACF decides whether to refer a patient for confirmatory testing using the radiograph, the patient's symptoms, and the CAD output [21].

C. Related Work

TB CAD models with lightweight memory and compute have been explored in previous studies, from LightTBNet at under a million parameters [49] to lightweight hybrid networks [50], building on edge-deployed chest X-ray models for related conditions [25], [26]. We do not benchmark LightTB-Net directly, because it was developed and validated for binary TB classification on Montgomery and Shenzhen-style datasets rather than the three-class TBX11K task evaluated here, so a fair head-to-head would require retraining it under this paper's protocol. Some existing commercial CAD systems require a dedicated CAD box or separate CAD laptop to support offline screening due to compute constraints [21].

Related literature has also explored models that generate natural-language descriptions of findings. In a research prototype, Siemens Healthineers paired a combined classification-localization model [12] with a domain-adapted LLM to generate CXR findings [11]. General medical vision-language models such as LLaVA-Med [13] and CXR-LLaVA [29] demonstrate image-conditioned radiology text generation, and recent surveys catalogue the rapid growth of vision-language models for medical report generation [30]. However, most work remains in general chest X-ray detection, with few specific to tuberculosis.

II. System Overview

This paper presents a lightweight TB screening pipeline that describes model findings in natural language, as shown in Figure 2. A CXR (DICOM or standard image format) is ingested from the screening unit's X-ray or uploaded directly, then scored by the TB classification module. For images flagged as TB, the object detection module identifies active and latent TB regions. The classification and detection outputs are passed to the report generation module, which produces a natural-language report of the findings based on radiologic report convention. All three stages are lightweight, for screening where no radiologist and often no GPU is present.

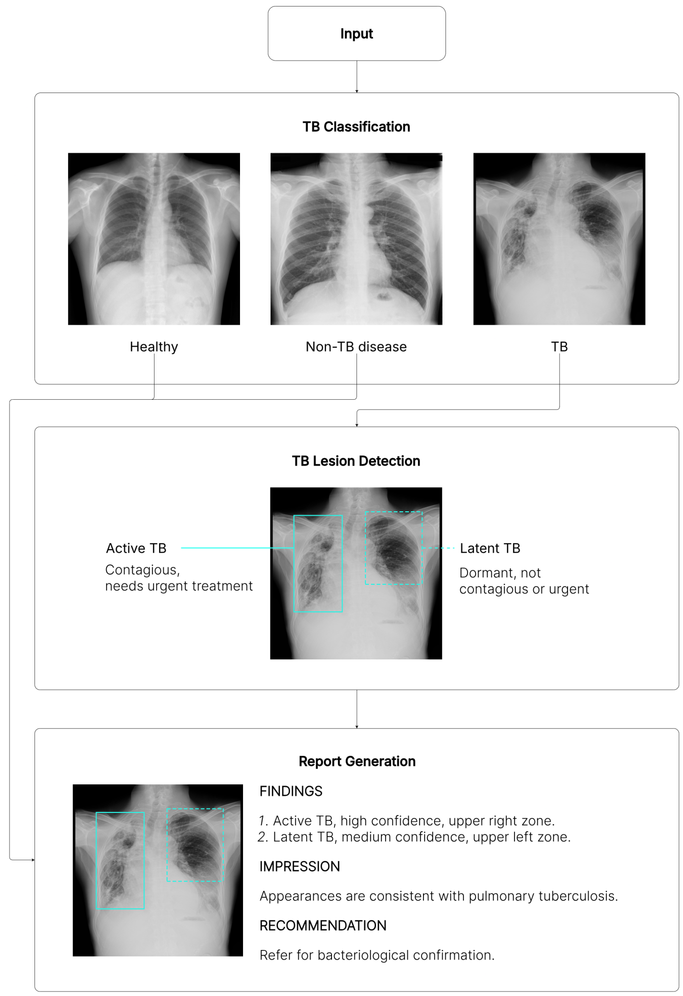

Fig. 2. Overview of the TB CAD pipeline: image classification, TB lesion detection, and report generation.

III. Tuberculosis Classification

The TB classification module flags each CXR as either healthy, sick non-TB, or TB as shown in Figure 2.

A. Dataset

Classification and object detection models are trained on a simplified redistribution[1] of TBX11K [39], a public CXR dataset with image-level labels (*healthy*, *sick non-TB*, *TB*) and bounding-box-level labels (*active-TB*, *latent-TB*).

TBX11K is selected because it is the largest public TB CXR dataset with a TB-specific sample large enough to train from scratch, expert bounding-box annotations, a sick non-TB class, and a demographic and class composition that reflect real-world clinical settings [40]. Its image-level labels are established by diagnostic microbiology and reviewed by experienced radiologists [40], the same microbiological reference standard used in the WHO 2025 product evaluation [7], rather than by image reading alone. TBX11K notes per-image patient age and sex, with most TB cases aged 20 to 60 and a male majority (72% of TB X-rays), which is representative of the general population and consistent with real-world clinical scenarios [40].

The official TBX11K test set is withheld for an online challenge, so we use only the public trainval images. Our results are therefore not directly comparable to SymFormer [40] or other challenge entries on the private test set.

Classification uses the image-level labels shown in Table I. The 8,400 public CXRs (3,800 healthy, 3,800 sick non-TB, 800 TB) are partitioned per class into train, validation, and test folds in a 70%/15%/15% ratio. All reported classification metrics are computed on the held-out test fold of 1,260 CXRs (570 healthy, 570 sick non-TB, 120 TB).

TABLE I. TBX11K DATASET COMPOSITION (IMAGE-LEVEL LABELS) FOR CLASSIFICATION

Class	Train	Val	Test	Total
Healthy	2,660	570	570	3,800
Sick non-TB	2,660	570	570	3,800
TB	560	120	120	800
Total	5,880	1,260	1,260	8,400

Per-class 70/15/15 train/validation/test split.

B. Data Preprocessing

Classification models are trained without preprocessing or data augmentation. TBX11K CXRs are acquired under a standardized posteroanterior protocol and are largely uniform in pose, so aggressive augmentation risks introducing distribution shift.

C. Model Architecture

We evaluate nine classifiers: six baselines (EfficientNet-B0 [44], MobileNetV3-Large [43], DenseNet-121 [42], ResNet-18, ResNet-50 [41], ConvNeXt-Tiny [45]) and three custom models (DraxNet, Drax-MobileNetV3-Large, FlipR). Full architecture of custom models are listed in Appendix A.

1) DraxNet: DraxNet (16.5M parameters) modifies a ResNet-18 backbone so the fourth stage consists of two residual blocks, each with a custom mixer (*DraxBlock*) inserted between its two convolutions. All other components are unchanged. *DraxBlock* consists of a ConvNeXt local convolutional branch followed by an optional self-attention branch, fused via stochastic-depth residuals. DraxNet, short for Dynamic Residual Attention eXchange, reserves the heavier mixer for the final, lowest-resolution stage, adding attention capacity with fewer parameters than a full attention backbone.

2) Drax-MobileNetV3-Large: Drax-MobileNetV3-Large (4.8M parameters) applies the DraxBlock to a parameter-efficient backbone. The MobileNetV3-Large feature extractor and head are unchanged, and a bottlenecked *DraxBlock* is inserted after the final feature stage. This tests whether the DraxBlock generalizes beyond a ResNet-style backbone.

3) FlipR: FlipR (2.8M parameters) inserts a *FlipR* block between the second and third stages of a ResNet-18 truncated to three stages, with the parameter-heavy fourth stage removed. The *FlipR* block computes the difference between each feature map and its horizontally flipped counterpart, smoothed by average pooling to account for imperfect anatomical symmetry, then broadcasts a gated mask across all channels. The block is inspired by SymAttention [40], which observes that bilateral asymmetry in CXRs is a key indicator of TB. FlipR tests whether a lightweight symmetry gate can match heavier attention approaches.

D. Model Training

All models are trained from scratch with a fixed random seed, batch size 16, 512×512 input, 100 epochs, Adam optimizer at a fixed learning rate of 10^{-3}, and unweighted cross-entropy loss. No hyperparameter tuning was done. The checkpoint with the lowest validation loss is retained and evaluated on the held-out test fold.

E. Results

Classification results are reported in Table II.

FlipR matches the top models on accuracy with the fewest parameters, though not the fewest multiply-adds. The top four models (FlipR, EfficientNet-B0, MobileNetV3-Large, Drax-MobileNetV3-Large) differ by within $\pm 1\%$ on accuracy and F1. FlipR uses the smallest parameter count (2.8M against 4.0 to 4.8M for the other three), consistent with the SymFormer [40] observation that bilateral asymmetry is a strong indicator of TB. Its parameter savings come from removing the parameter-heavy final stage of ResNet-18, which runs at low resolution and so contributes little compute. As a result FlipR carries the fewest parameters but not the fewest multiply-adds: at 7.4G it exceeds the depthwise-separable backbones (MobileNetV3-Large 1.2G, Drax-MobileNetV3-Large 1.3G, EfficientNet-B0 2.1G), which keep both parameters and compute low. Parameter count measures the storage and memory footprint, while multiply-adds measure compute, and the two favor different models.

All models exceed WHO triage targets on this test set, but not under the WHO 2025 evaluation protocol. Sensitivity ranges from 92.50 to 96.67%, above the 90%

[1]https://www.kaggle.com/datasets/vbookshelf/tbx11k-simplified

TABLE II. TUBERCULOSIS CLASSIFICATION RESULTS

Model	AUC_{TB}	ACC (%)	SENS (%)	SPEC (%)	PREC (%)	REC (%)	F1 (%)	Params	MultAdds
FlipR	0.9984	98.97	94.17	**99.91**	**99.01**	97.70	98.34	**2.8M**	7.4G
EfficientNet-B0	**0.9995**	98.89	95.00	99.65	98.29	97.87	98.07	4.0M	2.1G
MobileNetV3-Large	**0.9995**	**99.21**	**96.67**	99.82	98.97	**98.54**	**98.75**	4.2M	**1.2G**
Drax-MobileNetV3-Large	0.9989	98.81	95.00	99.65	98.23	97.81	98.02	4.8M	1.3G
DenseNet-121	0.9975	98.57	92.50	99.56	97.81	96.97	97.38	7.0M	15.0G
ResNet-18	0.9982	98.17	93.33	99.21	96.71	96.90	96.80	11.2M	9.5G
DraxNet	0.9988	98.41	95.00	99.47	97.51	97.51	97.51	16.5M	10.9G
ResNet-50	0.9978	97.70	92.50	99.47	96.95	96.33	96.63	23.5M	21.5G
ConvNeXt-Tiny	0.9952	97.14	95.00	97.98	93.63	96.58	94.97	27.8M	23.4G

AUC_{TB} is the area under the ROC curve for the TB class. Sensitivity is recall for the TB class. Specificity is recall for the non-TB class (healthy and sick non-TB combined). Precision, recall, and F1 are averaged across the three classes (healthy, sick non-TB, TB). Params is the trainable parameter count. MultAdds is the number of multiply-add operations for one 512×512 forward pass, and FLOPs are twice this.

sensitivity target, and specificity ranges from 97.98 to 99.91%, above the 70% specificity target of the WHO target product profile [2]. Sensitivity is more clinically important than specificity in screening, because a missed TB case risks untreated transmission while a false positive triggers only unnecessary confirmatory testing. MobileNetV3-Large achieves the highest sensitivity, missing only 3.3% of TB cases. These figures are not directly comparable to the WHO 2025 evaluation of approved products [7], for two reasons. First, the operating point differs: we report argmax sensitivity and specificity, whereas the WHO 2025 protocol fixes sensitivity at 90% and reports specificity there (with a clinical cutoff near 40%, not 70%) alongside a partial AUC over the 40 to 60% specificity range. Second, the test set differs in difficulty: although TBX11K shares the WHO microbiological reference standard, it is curated, single-source, and class-balanced, whereas the WHO libraries aggregate multiple regions and sources, and approved products reach only 26 to 56% specificity at 90% sensitivity on them. Reporting these WHO 2025 operating-point metrics on a comparable library is left to future work.

The high-specificity, lower-sensitivity profile is likely attributable to TBX11K rather than model architecture. All models display high specificity and lower sensitivity, consistent with results from models in related literature trained on TBX11K. The TBX11K dataset is constructed to reward high specificity because the common failure mode on other TB datasets is over-predicting TB and achieving near-zero specificity [40].

Lightweight models are preferred for TB screening on both performance and deployment grounds. The top four models (FlipR, EfficientNet-B0, MobileNetV3-Large, Drax-MobileNetV3-Large) match or exceed every larger model in accuracy and F1 with fewer parameters and fewer multiply-adds. Due to a modest TBX11K training set, a minority TB class, and no class-imbalance handling, larger networks appear to overfit the majority healthy and sick non-TB classes rather than learn discriminative TB features. TB screening also runs in resource-limited settings often without a dedicated GPU, where both the memory footprint set by parameter count and the inference latency set by multiply-adds matter. No single model minimizes both. FlipR has the fewest parameters, while the depthwise-separable backbones have the fewest multiply-adds, so the deployment choice depends on whether storage or compute is the tighter constraint on the target device.

IV. TUBERCULOSIS LESION DETECTION

We frame TB lesion detection as an object detection task: the module localizes each TB region on the radiograph as a bounding box and classifies it, supplying the region-level evidence that the report generation module later renders as text.

A. Dataset

Object detection uses the TBX11K bounding-box annotations, with each region labeled active TB or latent TB. Dataset composition is shown in Table IV.

Active TB denotes ongoing, transmissible infection requiring treatment. Latent TB denotes prior, inactive infection that is not contagious but may reactivate. As Figure 2 shows, active TB is more visually salient and latent TB is more subtle. A CXR with an image-level label of TB may contain bounding box labels of only active-TB lesions (630 images), only latent-TB lesions (139 images), or both (30 images). This composition is consistent with real-world conditions [39].

Differentiation between active and latent TB is relatively understudied in TB CAD. In routine reporting, a radiologist may not always state whether a lesion is suspected as active or latent TB explicitly, instead inferring it from radiographic pattern, with consolidation and cavitation suggesting active disease and fibrotic linear opacities suggesting healed disease, alongside clinical and laboratory correlation [4]. Current literature understudies whether existing commercial CAD systems can differentiate latent from active TB [21].

B. Model Architecture

We evaluate four lightweight detectors at 512px input: base YOLO26-n [46], two standard nano baselines YOLOv8n [47] and YOLO11n [48], and a custom variant in which YOLO26's default feature extractor is replaced by the DraxNet backbone used in classification. All detectors are trained from random initialization under the same detection head and training protocol. The DraxNet variant retains YOLO26's head, which isolates the backbone's effect relative to base YOLO26.

TABLE III. TUBERCULOSIS LESION DETECTION RESULTS

Model	**Class**	mAP_{50}	$\mathrm{mAP}_{50\text{-}95}$	**Params**	**MultAdds**
YOLO26	ATB	0.4511	0.2110	2.6M	2.0G
	LTB	0.1070	0.0410		
YOLO11n	ATB	0.5695	0.2550	2.6M	2.1G
	LTB	0.1595	**0.0759**		
YOLOv8n	ATB	**0.6120**	**0.2920**	3.2M	2.8G
	LTB	**0.1642**	0.0720		
YOLO26 + DraxNet	ATB	0.5865	0.2748	19.7M	12.7G
	LTB	0.1017	0.0310		

ATB = active TB, LTB = latent TB. Params and MultAdds are per model and identical for both classes. MultAdds is the number of multiply-add operations for one 512×512 forward pass, and FLOPs are twice this.

TABLE IV. TBX11K DATASET COMPOSITION (BOUNDING BOX LABELS) FOR LESION DETECTION

Annotation	**Train**	**Val**	**Total**
Active-TB boxes	724	248	972
Latent-TB boxes	178	61	239
Total boxes	902	309	1,211

Counts are individual bounding boxes, not images. The 799 TB images (599 train, 200 val) contain 1,211 boxes, since one image may hold several regions.

C. Model Training

Object detection uses the same training configuration as the classification models (Section III). As in classification, the best checkpoint (selected by mAP) is retained and used for evaluation.

D. Results

Detection results are reported in Table III.

Active TB is more likely to be detected than latent TB. All four detectors localize active TB several times more accurately than latent TB, roughly three to six times at mAP_{50} and three to nine times at $\mathrm{mAP}_{50\text{-}95}$. This gap reflects the TBX11K class imbalance (active-TB regions outnumber latent-TB regions roughly 4:1) and the subtle appearance of latent-TB lesions. The TBX11K reference reports the same pattern across all detection methods, attributing it to imbalance [40], and it is consistent with human readers distinguishing latent TB less reliably than active TB [5]. The bias toward active TB also aligns with clinical triage priorities, because active TB requires urgent treatment.

Compact stock detectors outperform the custom backbone on active TB. YOLOv8n attains the best active-TB localization at 0.6120 mAP_{50} and 0.2920 $\mathrm{mAP}_{50\text{-}95}$, exceeding both base YOLO26 and the DraxNet variant while using 3.2M parameters against DraxNet's 19.7M. YOLO11n is the next strongest on active TB at 2.6M parameters. The DraxNet backbone does improve over its own base: it raises active-TB mAP_{50} from 0.4511 to 0.5865 ($+0.1354$) and $\mathrm{mAP}_{50\text{-}95}$ from 0.2110 to 0.2748 ($+0.0638$), driven almost entirely by precision ($0.74 \rightarrow 0.82$) at unchanged recall (roughly 0.26), while slightly reducing latent-TB ($0.1070 \rightarrow 0.1017$ at mAP_{50}, $0.0410 \rightarrow 0.0310$ at $\mathrm{mAP}_{50\text{-}95}$). This controlled gain does not translate into an overall advantage, because the smaller stock detectors localize active TB more accurately at a fraction of the parameters. The DraxNet backbone also raises the detector to 19.7M parameters, far above the lightweight budget this pipeline targets for edge deployment. It is therefore not adopted for detection: a stock lightweight detector is both smaller and more accurate, and we use YOLOv8n as the detection model.

Recall, not precision, is the bottleneck. Both detectors localize confidently but find only about a quarter of the annotated TB regions. Figure 3 shows precision staying high while recall saturates near 0.26, and the latent-TB curve collapsing at low confidence for both detectors.

Localization is useful to confirm (rule-in) TB, but not to rule it out. The high-precision, low-recall profile indicates that a flagged lesion is likely correct and warrants focused radiologist review, but the model cannot reliably exclude TB because most annotated lesions are missed. Localization is therefore most reliable as supporting region-level evidence for the classifier and report generation modules, rather than as a standalone detector for TB. Clinicians using the model may reliably use it for rule-in diagnosis, not for rule-out diagnosis.

V. REPORT GENERATION

The report generation module turns the detector's output into a structured radiology-style report a reader can act on without interpreting a boxed region or heatmap. Example outputs are shown in Figures 2 and 4.

The module follows the two-stage detector-to-text decomposition of Danu et al. [11]. However, because the detector here emits a closed TB vocabulary rather than open findings, we replace their proprietary detector and fine-tuned language model with the public models of this paper and a deterministic module, reserving a generative model for future work that uses specific radiographic findings. For current use, template-based report generation suffices and does not require significant additional memory or compute for resource-constrained settings.

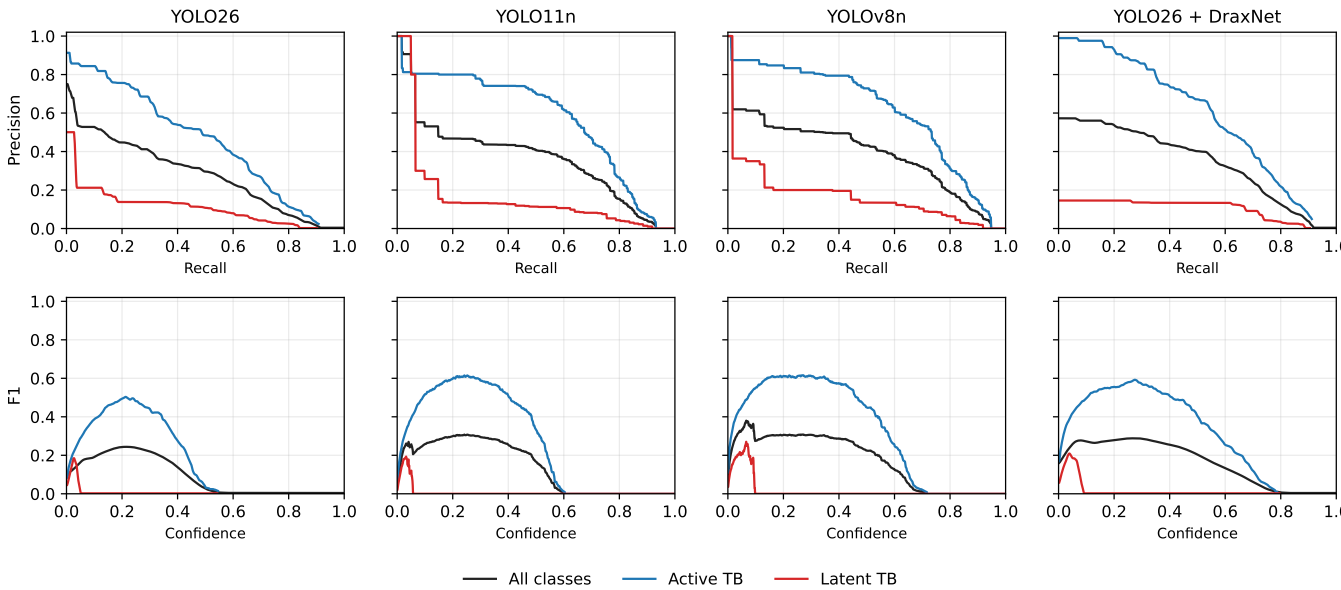

Fig. 3. Detection precision-recall (top) and F1-confidence (bottom) curves for the four detectors. The active-TB curve sits well above the latent-TB curve for every model, and the latent-TB curve collapses at low confidence, so localization functions as a high-precision rule-in cue rather than a standalone detector. The stock YOLOv8n and YOLO11n match or exceed the larger DraxNet variant on active TB.

The module has two stages: an adapter that converts raw detector output into a validated structured record, and a template-based report generator that renders that record as text. The second stage receives only a structured record of what the classification and object detection modules produced. This was designed to prioritize faithfulness to model findings, as image-conditioned vision-language models such as CXR-LLaVA [29] and LLaVA-Med [13], as well as related work using LLMs [11] may report findings the model detector never flagged.

A. Structured Adapter

The adapter maps classification and detection output to a fixed schema with a small, closed vocabulary. The image-level field carries the predicted label, the argmax over *healthy*, *sick non-TB*, and *TB*, together with its class probabilities. Each detected region carries three fields: its type (*active TB* or *latent TB*), a confidence band (*low*, *medium*, or *high*, thresholded at 0.5 and 0.8 on the detection score), and a location. The location is assigned by mapping the bounding-box centroid to one of nine cells in a 3×3 grid over the image (upper, middle, or lower row by left, center, or right column). An image with no detections yields an empty region list. The schema is validated before generation, so the report generator always receives well-formed input drawn from this vocabulary. The record states the type and coarse location of each region but no finer radiographic finding, because the detector labels regions only as active or latent TB.

All images are posteroanterior (PA) frontal scans. Hence, the standard radiological laterality convention is used, wherein the patient's right side appears on the image's left side and vice versa. For example, a lesion present in the image's upper left corner, is reported as an upper right lesion.

B. Template-Based Report Generation

Template-based report generation is a structured template over the closed schema. It renders the record as a short report in three labeled sections, mirroring the findings-to-impression structure of a radiology report [57]: a numbered FINDINGS list with one TB region per line including lesion type, model confidence band, and lesion zone, an IMPRESSION stating the screening result in clinical wording, and a RECOMMENDATION based on the predicted class, such as referral for bacteriological confirmation when TB is indicated.

The report describes only the detected TB regions and does not assess specific radiographic findings or cardiac, pleural, and skeletal observations. Due to lack of data, this is reserved for future work.

We chose a template over a generative model after prototyping both. An instruction-tuned language-model report generator (Phi-3.5-mini-instruct [56], run image-blind on the same record) was prototyped alongside this work, but the target is a short, fixed-format report with one line per region, which a template reproduces exactly over the closed vocabulary. The template therefore matches the generative output on faithfulness, readability, and the ordering of findings while removing any possibility of hallucination and the model inference cost, an advantage in the resource-limited settings this pipeline targets. A generative model is needed only for future work when the detector reports a richer, open set of interacting findings.

C. Faithfulness by Construction

By design, report generation is faithful by construction, defined as the generated text asserting nothing not present in the structured record. This follows the faithfulness-versus-factuality distinction drawn for abstractive summarization [51]

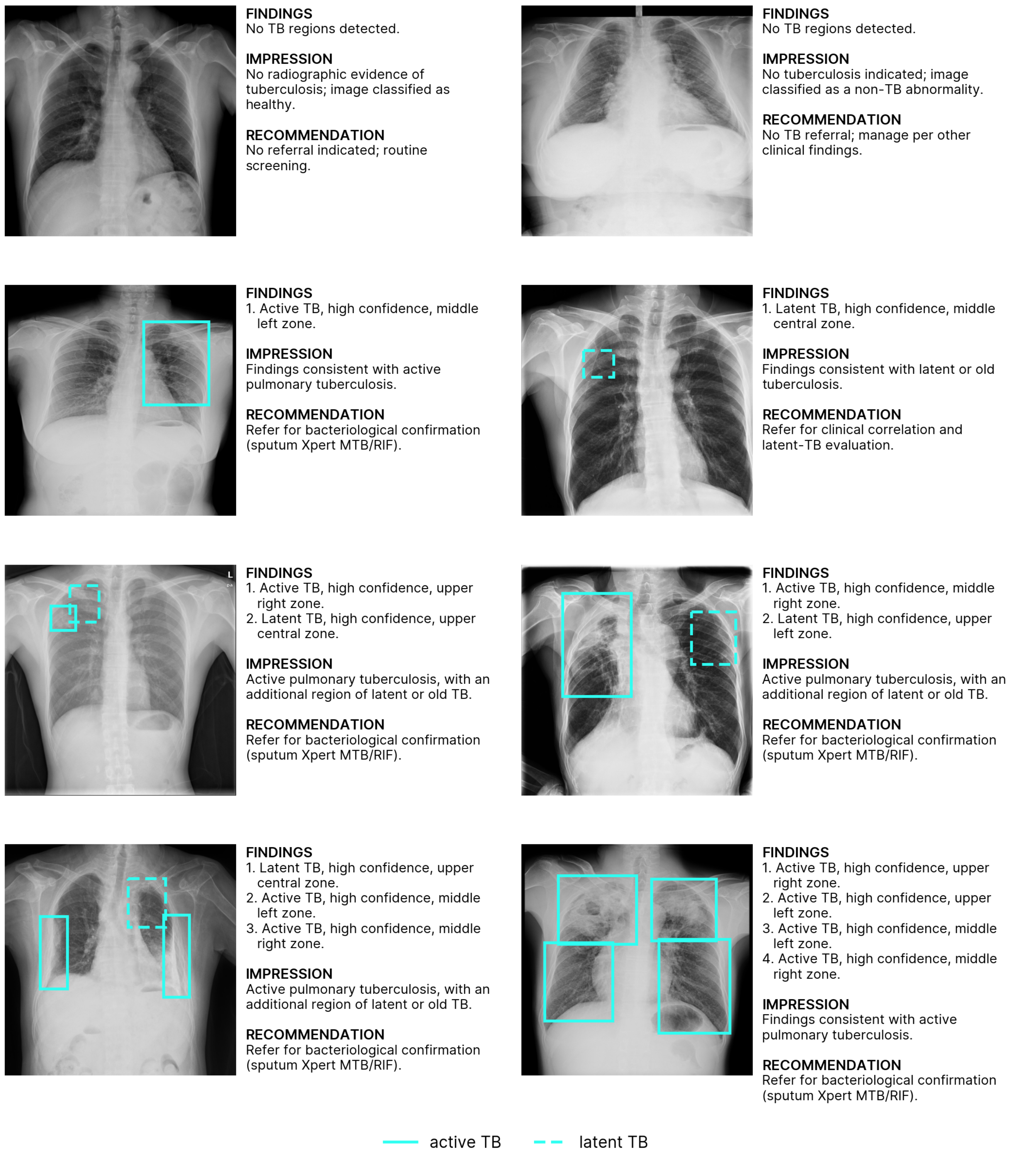

Fig. 4. Example generated reports on TBX11K scans.

and data-to-text evaluation [52], under which a generation is faithful if every entity it states is supported by the source, independent of whether the source is itself correct.

In previous experiments using Phi-3.5-mini-instruct for report generation, we evaluated the result for faithfulness with a deterministic verifier that parses a report for schema entities and flagging any the record does not contain: region type, location, or laterality absent from the record, a predicted label different from the record's, or an invented probability. The verifier follows an established approach in data-to-text generation, extracting the facts a generation asserts and checking them against the source records [53], and relates to the slot-error and semantic-accuracy metrics from dialogue generation that count output content not licensed by the input meaning representation [54], [55]. However, where those methods train an information-extraction or natural-language-inference model because the target text is open, the closed schema makes an exact deterministic match sufficient. Because the template renders deterministically over a validated closed schema, it cannot assert an entity absent from the record, so faithfulness holds by construction rather than as an empirical rate. The schema surface is small enough to confirm this exhaustively: every combination of the two region types, three confidence bands, and nine zones, together with the empty-region case for each class, parses with no unsupported entity. Since the generative approach does not provide additional benefit over the template, the generative model is dropped to avoid hallucination risk and reduce compute.

VI. Discussion

A. Deployment Feasibility

Laptops are typically used to run CAD systems in community-based ACF, resulting in limited compute. CAD is also increasingly deployed offline where network connectivity is unreliable or absent [23], [21]. This mode already operates at scale, with more than 430,000 people screened by ultraportable X-ray and CAD between 2021 and 2023 [24]. Models for community-based ACF must be able to run locally on a CPU, without a dedicated GPU or a network connection.

Our lightweight models fall at the parameter scale shown in related literature to run on this class of hardware. Lightweight chest X-ray networks have been deployed on edge devices and CPUs with low latency, including ResNet-18, MobileNet, and EfficientNet-B0 on an embedded board [25] and a compact classifier on edge devices [26], and for TB specifically LightTBNet reports rapid, low-memory prediction designed for embedded use [49]. FlipR (2.8M parameters), Drax-MobileNetV3-Large (4.8M), EfficientNet-B0 (4.0M), and MobileNetV3-Large (4.2M) are at or below the size of those deployed networks, so they should run on the same devices. Their multiply-add cost at the 512×512 input ranges from 1.2G for MobileNetV3-Large to 7.4G for FlipR, so the depthwise-separable backbones carry the smallest compute footprint as well as a small memory footprint, while FlipR trades higher compute for the fewest parameters. However, we did not conduct evaluation in real-world deployment to measure on-device latency or memory, and this remains future work.

B. Limitations

TB-specific samples are limited in public datasets. As of writing, TBX11K is the largest public chest X-ray dataset for tuberculosis with labeled bounding boxes, yet its TB subset remains small at 799 TB images and 1,211 TB lesion boxes. While general chest X-ray datasets exist, of these, TB-positive samples are few and insufficient to train from scratch. No public dataset contains TB-specific radiographs with reference radiographic reports, which is what a measurement of clinical correctness would require. This scarcity constrained the development and evaluation of report generation for clinical correctness.

The TB data scarcity limits how representative the evaluation population is. TBX11K is drawn from a single country, from patients presenting for hospital chest examination, and records no HIV status or prior-TB history. The WHO 2025 evaluation instead spans multiple regions, weighted toward high-burden African and South-East Asian settings, and reports lower CAD accuracy among women, people living with HIV, and those with a history of TB [7]. Reported performance here may therefore overstate what the pipeline would achieve in the higher-burden, HIV-prevalent populations that community active case finding targets, and evaluation on such populations is left to future work.

Although report generation is faithful by construction, faithfulness to the detector is not clinical correctness. A report that faithfully restates an incorrect detection is still wrong, and only reference-based or reader evaluation can report this. However, due to the lack of a TB-specific reference dataset, this is left to future work.

The system's scope is limited to screening of non-pediatric patients since the radiographic presentation of pulmonary TB is age-dependent: adult reactivation tuberculosis appears in the upper lung zones, whereas pediatric primary tuberculosis appears in the lower lung zones. Hence, detection patterns learned from a predominantly adult distribution do not transfer to children. This is consistent with the WHO endorsement of CAD as a human-reader alternative only for patients aged 15 years and older [6].

C. Future Work

Future work seeks to evaluate the deployment efficiency of the pipeline in real-world use to benchmark inference latency and memory on a target device, complementing the static parameter and multiply-add counts reported here.

Detection of full radiographic findings, such as cardiac size, pleural fluid, or non-TB abnormalities, as well as cardiac, pleural, and skeletal structures, are reserved for future work given additional data. Specific findings are more clinically meaningful for interpretation of the radiologic report, aligned with human reader interpretation of findings. This would also warrant use of a generative language model over a structured template for report generation, since synthesizing findings into a coherent report is no longer a fixed-format substitution but a genuine generation problem. This is demonstrated by recent chest X-ray systems that aggregate region-level findings into a report with an LLM [11], [58].

A reader study with radiologists and non-radiologist screeners is also recommended to evaluate the clinical correctness and coherence of the model-generated reports, particularly in comparison to human reader reports.

VII. Conclusion

TB CAD systems are highly accurate, however challenges remain in integration for use in community ACF with limited connectivity, compute, and with no radiologist present. This paper presented a lightweight TB screening pipeline that pairs lightweight detection models with deterministic, faithful-by-construction natural-language report generation of findings. The custom classifier, FlipR, achieves similar accuracy and F1 with the strongest baselines using the fewest parameters. For localization, a comparison of lightweight detectors shows that compact off-the-shelf models, led by YOLOv8n, localize active TB most accurately, and the DraxNet backbone improves active-TB precision over base YOLO26 but does not surpass the smaller stock detectors. Across all detectors, localization functions as a high-precision rule-in cue rather than a standalone detector. Report generation provides findings in natural language for interpretation of ACF field teams and radiographers for patient referral for presumptive TB. TB-specific samples were found to be highly limited in public datasets, which constrained the design and evaluation of components. A full evaluation of deployment feasibility, extended radiographic findings, and a reader study with radiologists is left to future work.

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Declaration on Generative AI

Generative AI tools were used to assist in drafting and preparing this manuscript. The conception of the research, methodology, and conclusions are the work of the authors.

Data and Code Availability

The data and source code for the classification, detection, and report-generation pipeline is publicly available at https://github.com/alive-ateneo/tb-community-screening-pipeline.

Appendix A
Architectures of Custom Models

TABLE V. FlipR Architecture (ResNet-18 Truncated to Three Stages)

Stage	Output	Params
Stem (conv 7×7, BN, ReLU, pool)	$64 \times H/4$	9,536
layer1 (2 basic blocks, 64)	$64 \times H/4$	147,968
layer2 (2 basic blocks, 128)	$128 \times H/8$	525,568
FlipR gate	$128 \times H/8$	129
layer3 (2 basic blocks, 256)	$256 \times H/16$	2,099,712
Head (avg-pool, FC 256→3)	3	771
Total		**2,783,684**

TABLE VI. DraxNet Architecture (ResNet-18, DraxBlock in Stage 4)

Stage	Output	Params
Stem (conv 7×7, BN, ReLU, pool)	$64 \times H/4$	9,536
layer1 (2 basic blocks, 64)	$64 \times H/4$	147,968
layer2 (2 basic blocks, 128)	$128 \times H/8$	525,568
layer3 (2 basic blocks, 256)	$256 \times H/16$	2,099,712
layer4 (2 DraxBlocks, 512)	$512 \times H/32$	13,699,072
Head (avg-pool, FC 512→3)	3	1,539
Total		**16,483,395**

TABLE VII. Drax-MobileNetV3-Large Architecture

Stage	Output	Params
MobileNetV3-Large features	$960 \times H/32$	2,971,952
DraxBlock adapter (bottleneck 160)	$960 \times H/32$	575,200
Head (avg-pool, classifier 960→1280→3)	3	1,233,923
Total		**4,781,075**

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